

Count Each Person at Their Changing Pace: Multi-Person Tempo-Adaptive Repetition Counting

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Abstract

DRAFT — RESULTS PENDING

*Multi-person repetitive action counting (MRAC) must assign repetitions to the correct person even when several people act asynchronously and each person accelerates, decelerates, pauses, or disappears behind an occluder. The MultiCounter family jointly learns detection, association, and counting from annotated RGB videos, while label-efficient pose counters are primarily developed for a single dominant actor. We study the complementary setting in which a counting model receives identity-indexed pose tracks but uses no person-wise count or period annotation during its intended counting training. Our proposed design shares one pose encoder and a bank of tempo experts across people while keeping temporal state separate for every identity. Within overlapping windows, masked period evidence drives a soft fast-medium-slow router. Expert responses are fused by normalized overlap-add, after which each identity is decoded once, removing the direct arithmetic double-counting path created by summing independent windows. The primary output is a person-wise count vector rather than only a scene total. **Experimental evidence is pending synchronization with the collaborator's frozen implementation and audited MultiRep runs.** The current manuscript is an evidence-gated pre-results specification: exact auxiliary losses, frontend choices, and every empirical statement remain blocked until they are reconciled with frozen code and machine-readable artifacts.*

1. Introduction

Repetitive motion is a useful measurement primitive. Exercise coaching, rehabilitation, industrial monitoring, sports analysis, and human-robot interaction all require more than recognizing that an action repeats: a system must determine how often each participant repeats it. In an uncontrolled scene, however, people need not move together. One person

may slow down while another accelerates, a third may pause, and tracks may contain gaps caused by occlusion. A single response curve for the whole frame can therefore mix identities and incompatible time scales, producing a plausible scene total while assigning the wrong count to each person.

Repetition counting has progressed from online local cycle estimation [4] and motion-frequency analysis [8] to learned temporal self-similarity, density regression, and dynamic-query models [1, 3, 6]. ESCounts studies exemplar-conditioned counting and predicts one target density/count stream [9]. SkimFocusNet studies exemplar-selected multiple action types [15]. These advances largely assume one counted stream. MultiCounter formalizes multi-instance repetitive action counting and jointly predicts people, identities, periodic intervals, and counts [10]; MultiCounter+ strengthens temporal identity consistency and long/short-period awareness [11]. Consequently, multi-person, person-wise, asynchronous, and variable-speed counting are established problems rather than novelty claims of this paper.

The remaining opportunity lies at the intersection of supervision, modality, and temporal granularity. Existing MRAC systems are trained as supervised RGB pipelines, whereas pose-based and label-efficient counting has been developed mostly for a single instance. PoseRAC exploits salient poses in an open-set, zero-shot framework [13]; D²-STX combines RGB and pose streams [12]; and PAMS learns pose representations through period-adaptive multi-scale consistency without repetition annotations [2]. PAMS also evaluates robustness under uniform global speed resampling, but it does not separate simultaneous identities. Conversely, MRAC identity mechanisms do not directly provide an annotation-efficient, pose-driven counter whose tempo assignment can change from window to window along one person's track.

We study that constrained setting. An upstream perception system supplies identity-indexed pose tracks and validity masks. The counting model shares its encoder and tempo experts across people, but maintains independent buffers, routing weights, and output streams for each identity. A masked local period estimate selects a soft mixture of fast, medium, and slow experts within every overlapping win-

075 dow. Instead of counting windows and adding their integer
076 outputs, the model stitches continuous responses with nor-
077 malized overlap-add and performs one final peak extraction
078 per track. This ordering makes the window boundary explicit
079 and removes the direct arithmetic path by which one repeti-
080 tion can be independently counted in two overlapping win-
081 dows; it does not guarantee immunity to phase-misaligned
082 responses.

083 Figure 1 contrasts the target output with a mixed single-
084 stream counter. The formulation does not assume that track-
085 ing errors have been solved: predicted-track experiments
086 must expose their effect, while oracle-track experiments iso-
087 late the counting module. It also does not call the whole
088 system free of all labels. A detector, pose estimator, tracker,
089 or backbone may have external supervision; the narrower
090 intended property is that the counting objective consumes
091 no person-wise repetition count or period labels.

092 The paper makes three falsifiable, currently evidence-
093 gated contributions:

- 094 • We formulate a pose-driven MRAC training setting whose
095 intended counting loss uses no person-wise count or pe-
096 riod annotations, while explicitly accounting for externally
097 supervised perception components. This supervision state-
098 ment must be verified against the frozen training graph
099 before submission.
- 100 • We propose track-indexed, window-level tempo routing
101 with shared parameters and per-track state, followed by
102 response-domain overlap-add and one track-level decod-
103 ing pass. Its benefit must be separated from the effects of
104 tracking and the PAMS representation through controlled
105 ablations.
- 106 • We establish an evaluation contract that pairs the official
107 MultiRep counting metrics with oracle/predicted-track pro-
108 tocols, tracking diagnostics, non-stationary-tempo stress
109 tests, and person-count scaling. Every final table cell must
110 be generated from a frozen result manifest.

111 This pre-results draft freezes the scientific story and its
112 evidence requirements; it does not assert that the multi-
113 person modules are already implemented or that the pro-
114 posed method outperforms any baseline.

115 2. Related Work

116 **Single-instance counting and tempo modeling.** Early
117 learned counters estimated a local cycle length in an
118 online window [4], while motion-centric methods used
119 time-varying flow differentials and wavelets to address
120 non-stationary repetition [8]. RepNet used temporal self-
121 similarity as an intermediate representation bottleneck and
122 introduced Countix as an in-the-wild benchmark for class-
123 agnostic repetition counting [1]. TransRAC regresses a den-
124 sity map from multi-scale temporal correlations [3], and
125 DeTRC represents cycles with dynamic action queries under
126 a single-primary-action assumption [6]. ESCounts

127 uses exemplar-conditioned correspondence during training
128 and learns a latent used for exemplar-free, zero-shot infer-
129 ence [9]; SkimFocusNet combines coarse contextual guid-
130 ance with fine-grained views. Multi-RepCount evaluates
131 exemplar-specified counting of one target action category
132 in videos containing multiple repetitive-action categories,
133 whereas MultiRep evaluates per-person counting for simul-
134 taneous human instances [10, 15].

Pose-based and label-efficient counting. Pose represen-
135 tations reduce appearance dependence and expose articu-
136 lated motion directly. PoseRAC models actions through
137 salient poses, uses foundation generative models to synthe-
138 size pose-conditioned training data, and incorporates CLIP
139 text embeddings for open-set counting, enabling zero-shot
140 prediction without real benchmark training samples [13];
141 zero-shot prediction should not be conflated with a self-
142 supervised training objective. D²-STX instead fuses RGB
143 and pose through decoupled spatial and temporal cross-
144 attention [12]. PAMS learns from unlabeled skeleton se-
145 quences using period-adaptive multi-scale temporal consis-
146 tency and combines inference-time counting experts [2]. Its
147 expert consensus is not a learned fast-medium-slow mixture,
148 and its single-stream formulation does not preserve multi-
149 ple identities. Our intended use of pose and PAMS-TCC is
150 therefore a foundation, not a novelty claim; the proposed
151 distinction is track-indexed local routing in MRAC under
152 a no-count-label objective. Local routing and normalized
153 overlap-add are generic operations and are not claimed as
154 independent inventions; any defensible novelty must come
155 from their audited role in the stated MRAC supervision and
156 non-stationary-tempo setting.

Multi-instance repetition counting. MultiCounter de-
158 fines MRAC and introduces MultiRep, learning detection,
159 association, period localization, and person-wise counting
160 jointly [10]. MultiCounter+ improves robustness and ef-
161 ficiency with a dynamically updated Siamese query and
162 anchor-guided similarity reconstruction that represents long
163 and short periods [11]. These systems already address asyn-
164 chronous people, variable speeds, pauses, and identity conti-
165 nuity. Our proposed setting differs along a narrower axis: an
166 upstream tracker supplies masked pose tracks, the counting
167 objective is intended to avoid person-wise count and period
168 labels, and tempo specialization varies across local windows
169 of each identity. This modular design is not described as
170 end-to-end, and any comparison to native MRAC systems
171 must disclose the different input and supervision protocols.
172 Appendix 5 makes these boundaries explicit; it also prevents
173 generic local routing or overlap-add from being presented as
174 a standalone novelty.

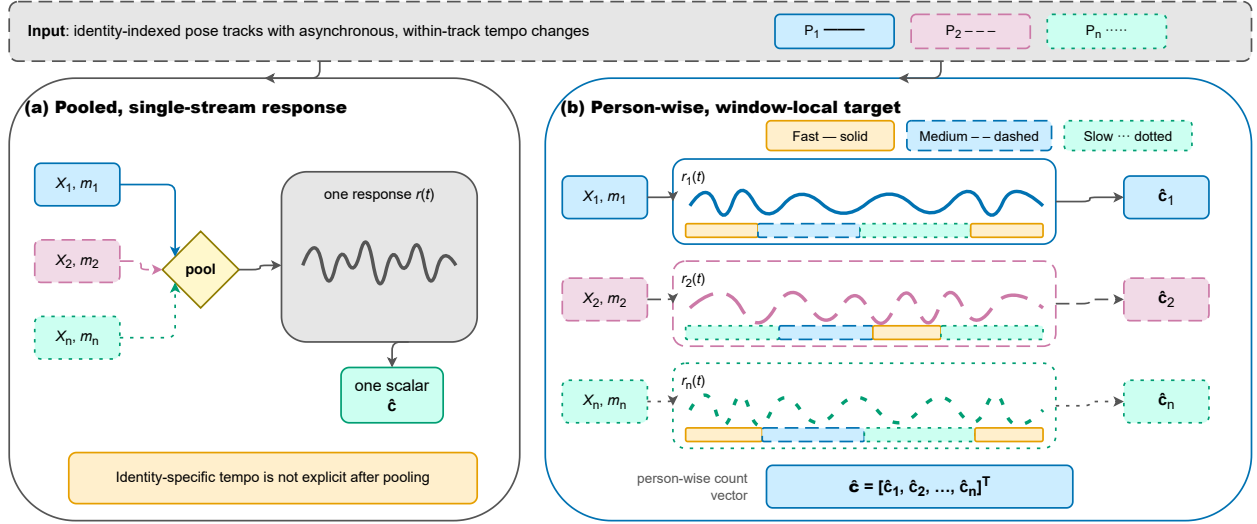


Figure 1. **Target task and output.** The left panel is a naïve pooled single-stream failure mode, not MultiCounter or MultiCounter+. A frame-level response can mix people who move at different and changing rates (left). We instead retain identity-indexed masked pose tracks, estimate tempo locally within each track, and return a person-wise count vector (right). The scene total is only a derived diagnostic. This is a problem illustration, not result evidence.

3. Method

Status boundary. This section is a provisional method contract for synchronizing the paper with the collaborator’s frozen implementation. It defines the intended data flow and the boundary between upstream perception and counting. Components marked **SYNC-REQUIRED** are not asserted as implemented or validated. If the frozen code differs from this contract, the equations and prose must change to match the code. The narrow supervision statement used throughout this draft is “without person-wise count or period supervision in the intended counting objective”; it does not describe the supervision used by the upstream detector, pose estimator, tracker, or any pretrained representation.

3.1. Problem Formulation and Overview

Given a video, an upstream detector, pose estimator, and multi-object tracker produce N identity-indexed pose tracks. Track i consists of joint coordinates and a validity mask

$$\begin{aligned} \mathbf{X}_i &= [\mathbf{x}_{i1}, \dots, \mathbf{x}_{iT_i}] \in \mathbb{R}^{T_i \times J \times C} \\ \mathbf{m}_i &= [m_{i1}, \dots, m_{iT_i}] \in \{0, 1\}^{T_i} \end{aligned} \quad (1)$$

Here T_i is the temporal extent, J the joint count, C the coordinate dimension, and $m_{it} = 1$ denotes a usable pose at time t . Missing observations keep their original time indices and are masked rather than deleted. An upstream frontend may additionally expose joint-level visibility or confidence, but its deterministic reduction to m_{it} must be recorded in the synchronized preprocessing contract. The

primary prediction is

$$\hat{\mathbf{c}} = [\hat{c}_1, \dots, \hat{c}_N]^\top \in \mathbb{N}_0^N \quad (2)$$

The optional total $\sum_i \hat{c}_i$ is a diagnostic rather than the task definition. Under oracle tracks, index i denotes a ground-truth person and Equation 2 is person-wise. Under predicted tracks it denotes a tracker output, so the raw vector is track-wise: fragmentation may create several entries for one person and a merge may combine people. Person-wise predicted-track metrics are defined only after the frozen evaluator’s matching step. The counter cannot undo an identity switch already embedded in its input; Section 4 therefore separates oracle and predicted tracks.

The design in Figure 2 uses a shared pose encoder and a shared bank of fast, medium, and slow experts. “Person-specific” means that observations, masks, temporal buffers, routing variables, and outputs are indexed by identity; it does not mean that a separate parameter set is instantiated for every person or that an identity embedding conditions the network. Computation is track-indexed and otherwise independent across people. Each track is processed in overlapping windows, whose continuous responses are fused before any discrete counting decision.

3.2. Identity-Conditioned Local Encoding

A single track can accelerate, decelerate, pause, or contain observation gaps, so we do not assign one speed to the whole person. Let L and S denote window length and stride, and let $s_{i\ell}$ be the start of window ℓ . Its local input at $u \in \{1, \dots, L\}$

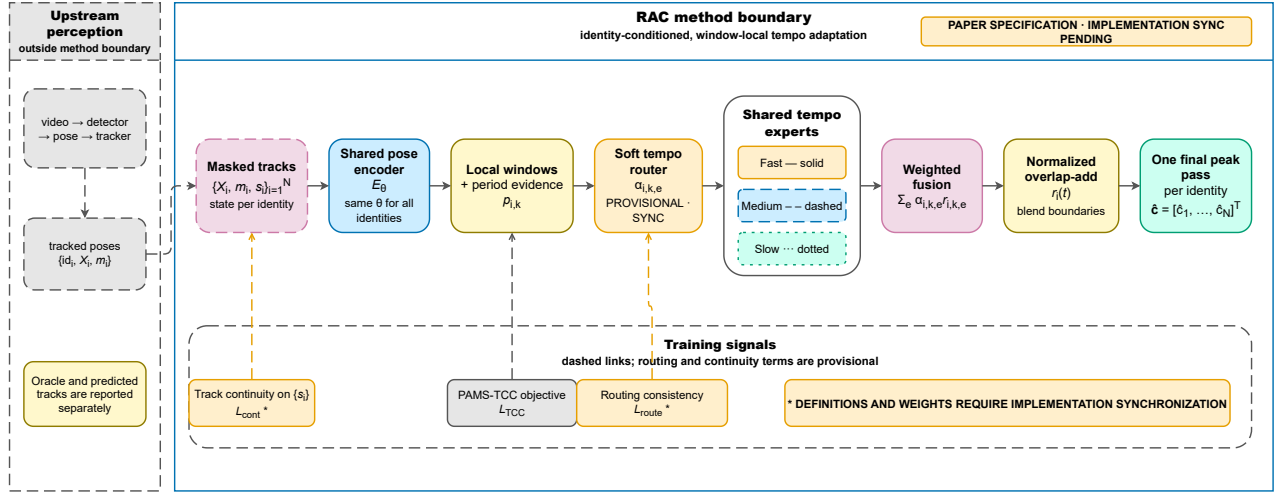


Figure 2. **Provisional method contract.** The upstream detector, pose estimator, and tracker are outside the counting boundary. Within it, parameters are shared across identities while masks, state, local tempo, and output responses remain identity-indexed. Window responses are fused by normalized overlap-add before one final decoding pass per track. Dashed elements require synchronization with frozen training code.

is

$$\mathbf{x}_{ilu} = \mathbf{x}_{i,s_{il}+u-1}, \quad m_{ilu} = m_{i,s_{il}+u-1} \quad (3)$$

Out-of-range positions are padded with zero mask values. A shared encoder E_θ maps each masked window to embeddings

$$\mathbf{Z}_{il} = E_\theta(\mathbf{X}_{il}, \mathbf{m}_{il}) = [\mathbf{z}_{il1}, \dots, \mathbf{z}_{ilL}] \in \mathbb{R}^{L \times d} \quad (4)$$

Invalid positions are excluded from aggregation, period evidence, and counting responses. Independent application of the same encoder places people in one representation space without pooling their temporal state; d denotes the embedding dimension.

3.3. Local Period Evidence and Soft Routing

We derive routing evidence within each window. With a fixed numerical stabilizer $\epsilon > 0$, the valid-frame mean is

$$\bar{\mathbf{z}}_{il} = \frac{\sum_{u=1}^L m_{ilu} \mathbf{z}_{ilu}}{\sum_{u=1}^L m_{ilu} + \epsilon} \quad (5)$$

On the uniform lag grid $\delta \in \{0, \dots, L-1\}$, let $n_{il}(\delta) = \sum_{u=1}^{L-\delta} m_{ilu} m_{il,u+\delta}$. Write $\tilde{\mathbf{z}}_{ilu} = \mathbf{z}_{ilu} - \bar{\mathbf{z}}_{il}$. We define a mask-normalized, non-circular autocorrelation and deterministically zero-fill unsupported grid locations

$$Q_{il}(\delta) = \sum_{u=1}^{L-\delta} m_{ilu} m_{il,u+\delta} \langle \tilde{\mathbf{z}}_{ilu}, \tilde{\mathbf{z}}_{il,u+\delta} \rangle \quad (6)$$

$$A_{il}(\delta) = \frac{\mathbf{1}[n_{il}(\delta) \geq n_{\min}] Q_{il}(\delta)}{n_{il}(\delta) + \epsilon}$$

The support mask is retained separately so a zero-filled lag is never treated as observed evidence. For a deterministic taper h , let $\mathcal{F}$ be the nonzero frequency bins in cycles per frame whose reciprocal periods lie inside the declared range. We use the magnitude of the tapered autocorrelation transform as a routing score, not as a calibrated physical power spectral density

$$P_{il}(f) = |\text{FFT}(h \odot A_{il})_f|$$

$$f_{il}^* = \arg \max_{f \in \mathcal{F}} P_{il}(f), \quad \hat{p}_{il} = 1/f_{il}^* \quad (7)$$

$$\gamma_{il} = \frac{P_{il}(f_{il}^*)}{\sum_{f \in \mathcal{F}} P_{il}(f) + \epsilon} \quad (8)$$

Here $\hat{p}_{il}$ is measured in frames and is converted to seconds only with the manifest-bound frame rate. The contract stops the direct gradient from this evidence branch into the encoder; encoder updates can still change routing evidence on later optimization steps. A router maps the descriptor $\mathbf{q}_{il} = [\log \hat{p}_{il}, \gamma_{il}]^T$ to a probability simplex over shared experts $\mathcal{K} = \{f, m, s\}$. Let π_0 be a fixed fallback distribution for a supported but low-confidence window

$$\tilde{\mathbf{g}}_{il} = \text{softmax}(R_\psi(\mathbf{q}_{il}))$$

$$\mathbf{g}_{il} = \gamma_{il} \tilde{\mathbf{g}}_{il} + (1 - \gamma_{il}) \pi_0 \quad (9)$$

$$\sum_{k \in \mathcal{K}} g_{ilk} = 1$$

An eligibility flag $v_{il} \in \{0, 1\}$ is one only when the window meets the frozen minimum valid-pair, minimum-cycle,

activity, and spectral-prominence rules. If $v_{i\ell} = 0$, the implementation assigns a tensor-shape-only reference period, sets $\gamma_{i\ell} = 0$, and forces the window response to zero; it does not route a pause or unsupported window through a counting expert. The exact activity statistic, thresholds, harmonic-selection rule, and whether a learned null expert replaces this deterministic gate remain synchronization-required. Because $g_{i\ell}$ is local, it may change between adjacent windows of one identity. Soft assignment also lets an ambiguous transition combine neighboring experts rather than commit to a hard tempo bin. The minimum support, confidence calibration, and whether π_0 is uniform or period-conditioned are synchronization-required. Expert-use histograms, entropy, and specialization curves must diagnose routing collapse.

3.4. Expert Fusion and Boundary Handling

Each expert F_{ϕ_k} predicts a nonnegative response over one masked window. Parameters ϕ_k are shared across identities, while each response remains identity-indexed

$$\mathbf{r}_{i\ell}^{(k)} = F_{\phi_k}(\mathbf{Z}_{i\ell}, \mathbf{m}_{i\ell}, \hat{p}_{i\ell}) \in \mathbb{R}_{\geq 0}^L$$

$$r_{i\ell u} = v_{i\ell} m_{i\ell u} \sum_{k \in \mathcal{K}} g_{i\ell k} r_{i\ell u}^{(k)} \quad (10)$$

Summing integer window counts creates a direct arithmetic double-counting path for cycles that straddle an overlap. We instead reconstruct a single response for each identity. With a taper $a_u \geq a_{\min} > 0$ on every covered frame and $\Lambda_i(t) = \{(\ell, u) : t = s_{i\ell} + u - 1\}$, normalized overlap-add gives

$$R_{it} = m_{it} \frac{\sum_{(\ell, u) \in \Lambda_i(t)} a_u r_{i\ell u}}{\sum_{(\ell, u) \in \Lambda_i(t)} a_u m_{i\ell u} + \epsilon} \quad (11)$$

The ratio replaces the direct coverage-count scaling of an unnormalized sum with an ϵ -stabilized valid-taper weighted average and excludes padding from the denominator. Its amplitude can still vary with the responses being averaged and retains a small support-dependent bias when $\epsilon > 0$, which the evaluation contract audits explicitly. Only then does a track-level decoder $\mathcal{D}$ extract repetition events

$$\hat{\mathcal{P}}_i = \mathcal{D}(\mathbf{R}_i, \mathbf{m}_i, \{\hat{p}_{i\ell}, \gamma_{i\ell}\}_{\ell}), \quad \hat{c}_i = |\hat{\mathcal{P}}_i| \quad (12)$$

Thus overlapping windows never contribute separately rounded counts. Fusion does not by itself guarantee a single peak when window responses are phase-misaligned; boundary-straddling tests must measure whether the fused decoder reduces such errors. The frozen implementation must determine the taper, coverage rule, peak threshold, period aggregation, and minimum-distance rule. Constant motion, pauses, contiguous gaps, alternating masks, harmonic ambiguity, and boundary-straddling cycles are mandatory unit-test cases. The mask-normalized estimator in Equation 6 is not pairwise variance-normalized, so corruption tests must also quantify mask-induced spectral artifacts.

For Period-mAP evaluation, the event set must expose more than its cardinality. The synchronized adapter must export each proposal as

$$\hat{c}_{ij} = (i, \hat{\tau}_{ij}^s, \hat{\tau}_{ij}^e, \hat{s}_{ij}, \hat{\mathcal{B}}_{ij}) \quad (13)$$

where the tuple contains its track identity, temporal bounds, confidence, and the spatial/track support required by the frozen official evaluator. The count remains $\hat{c}_i = |\hat{\mathcal{P}}_i|$. Exact interval construction, score calibration, and evaluator matching remain blocked until that adapter is bound to code and tests.

3.5. Training and Synchronization Boundary

The intended representation objective builds on PAMS-TCC [2], which uses temporally adjacent and cyclically corresponding positives across temporal scales, together with negatives from other temporal locations, other videos, and different clusters. At the method-contract level, training is represented by

$$\mathcal{L} = \mathcal{L}_{\text{PAMS-TCC}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}} + \lambda_{\text{track}} \mathcal{L}_{\text{track}} \quad (14)$$

The reserved routing term should transform assignments equivariantly under declared tempo changes rather than force invariance to them. By default, the reserved track term may connect observations only across gaps that retain one supplied track identity. It does not join fragments carrying different tracker IDs; any such re-association would be a separate module with its own signal, supervision disclosure, and evaluation. These are responsibilities, not recovered loss definitions.

SYNC-REQUIRED. Equation 14 is not yet a complete trainable algorithm: it does not specify what trains expert responses or the decoder, how experts specialize, how router collapse is prevented, or how scores are calibrated. Synchronization must recover the exact architectures, response targets or self-supervised signals, pseudo-target provenance, formulas, sample construction, weights, gradient paths, front-end, track normalization, period range, window schedule, fallback route, and decoding settings. Every use of count, period, or identity annotations in training, tuning, calibration, and checkpoint selection must be disclosed. No contribution or result claim depending on these items is admissible before source anchors, tests, and ablation evidence exist.

4. Experiments

Pre-results status. This section fixes the evaluation questions, protocols, and table schemas; it contains no new-method measurement or empirical conclusion. Red entries are controlled evidence fields. Historical UCFRep dev84 diagnostics, proxy readouts, and runs involving an untrained random Period Head are ineligible for every table below.

4.1. Evaluation Contract

Data and integrity. The primary benchmark will be the exact MultiRep release **PENDING** under its official **PENDING** protocol. The original MultiCounter paper specifies 1,157 synthetic videos, 52,590 periodic events, and a 7:2:1 train/validation/test split, while the current MultiCounter+ repository reports the same size and provides a download without a versioned release identifier or checksums [10, 11]. The final manifest must therefore bind the release identifier, download source and time, video/annotation/split SHA-256 digests, evaluator revision, and preprocessing fingerprint. A run is table-ineligible if any binding is missing or if incompatible artifacts are mixed. Before model comparison, a frozen dataset characterization report must give split-level counts of videos, ground-truth people, events, active-person levels, within-track tempo dispersion, detected accelerations/decelerations, pauses, observation gaps, and identities with enough cycles for every declared window length. Stratum definitions and counts are fixed before looking at method errors.

Oracle and predicted tracks. The oracle-track protocol supplies official person identities and spatial support, isolating repetition counting from detection and association. The predicted-track protocol uses one frozen detector–pose–tracker frontend for all compatible track-conditioned counters. Native joint MRAC systems retain their audited official pipelines and appear in a separate contextual block, not as if their inputs were controlled. Against ground-truth identities and spatial support, HOTA and its DetA, AssA, and LocA components [5] diagnose predicted-track detection, association, and localization quality. IDF1 and ID-switch counts [7] provide complementary identity-duration and temporal-association diagnostics. When the same frozen tracks feed several counters, these tracking values must be identical and cannot be credited as counting gains.

Evaluator-ready outputs. Each counter adapter must emit the scored event tuples in Equation 13, the raw per-track count vector, and the exact spatial support expected by the pinned evaluator. Oracle tracks retain official identity support. Predicted tracks remain separate through evaluation: the manifest-bound adapter must specify one deterministic matching policy and how unmatched, duplicated, fragmented, merged, and identity-switched tracks become false positives or false negatives. No post-hoc fragment joining is allowed unless it is a separately audited method component. Adapter source, tests, and per-instance outputs are required before Period-mAP or matched person-wise count metrics become table-eligible. Predicted-track outputs are called track-wise until this matching is applied.

Metrics and uncertainty. Using the frozen official evaluator, we will report Period-mAP, AP50, AP75, AvgMAE, and AvgOBO, together with a matched person-count error diagnostic after the evaluator’s frozen matching step. Period-mAP under common predicted tracks is the confirmatory metric; the primary contrast is our method minus track-conditioned PAMS on identical videos and frontend outputs. A claim of improvement requires the lower endpoint of its paired 95% interval to be above zero. Native-pipeline rows are descriptive and cannot establish non-inferiority or superiority under a common-input protocol.

Every reproducible method uses at least three pre-registered seeds, reported as mean and between-seed sample standard deviation. The confirmatory interval uses one pre-declared paired hierarchical bootstrap: each replicate resamples completed seed pairs and then videos within each sampled seed, retaining every identity of a sampled video and recomputing Period-mAP from event records rather than averaging per-video AP. Its point estimate is the mean of paired seed-level differences. The manifest fixes the resample count, random seed, failed-run policy, and exact numbers of completed seeds, videos, people, and events. Secondary ablations are exploratory, with family-wise Holm correction if formal significance statements are made. Official single-checkpoint numbers remain explicitly marked and are excluded from variance and significance claims. Test annotations are not used for tuning, checkpoint selection, routing calibration, or early stopping.

The numerical stabilizer ϵ is fixed before evaluation in the units of each denominator. A sensitivity check spans representable smaller and larger values and reports changes in response amplitude and count metrics by valid support; it must expose, rather than hide, any coverage dependence introduced by Equations 5 and 11.

Baselines and supervision. The mandatory native baselines are MultiCounter and MultiCounter+ [10, 11]. Under the common track interface we adapt RepNet [1], TransRAC [3], PoseRAC [13], and PAMS [2]; DeTRC [6] and D²-STX [12] are candidate baselines and enter only if runnable public implementations are available and their output semantics pass the experiment audit. Tables disclose input modality, training labels, track source, run count, adapter code, crop/pose construction, training split, tuning budget, pretrained checkpoint, and output conversion. Comparisons across supervision regimes are accuracy–annotation trade-offs, not claims of identical training information.

4.2. Primary Questions

The evaluation is organized around four questions. First, how does a no-count-label pose counter compare on person-wise oracle-track MRAC and on matched predicted tracks? Second, does local routing help beyond a track-conditioned

PAMS base when tempo changes within one identity? Third, does response-domain overlap-add reduce boundary errors relative to summing independently decoded windows? Fourth, how do tracking errors, active-person count, and upstream preprocessing affect robustness and efficiency? Each question has a named table or figure and an associated claim that remains withheld until the evidence audit passes. The mechanism study must also compare against running PAMS independently per track, sliding-window PAMS without a learned router, deterministic tempo bins, non-overlapping and center-crop decoding, uniform response averaging, and phase-aligned fusion.

The quantitative interpretation of Table 1 remains withheld until every displayed binding passes the empirical audits.

The ablation interpretation remains withheld until all rows share one frozen protocol and have the required paired records. The full mechanism study also removes each component one at a time, crosses local-tempo estimation, routing, and stitching in a small factorial design, matches single- and multi-expert capacity, tests experts with and without direct $\hat{p}$, and reports route occupancy, entropy, collapse, and specialization by tempo stratum.

The appendix pre-registers router variants, expert counts, local-versus-global tempo, shared batched execution versus independent same-checkpoint replicas, boundary and continuity ablations, tempo/occlusion/identity-switch stress, cross-dataset transfer, and latency, memory, FLOPs, and throughput as the number of active identities grows. All plots and tables must be generated from frozen JSON, CSV, or Parquet records; manually transcribed values are inadmissible.

5. Conclusion and Limitations

We specify a pose-driven MRAC counter for people whose pace changes within a track. The central design separates identity state while sharing parameters, routes overlapping windows from local period evidence, fuses continuous responses by normalized overlap-add, and decodes each track once. The setting does not remove all annotation and does not jointly optimize perception and counting; the counting objective is intended to avoid person-wise count and period labels, while upstream perception may use external supervision.

This draft has three material limitations. The current local checkout contains only a single-person PAMS reproduction; the multi-person data path, router, continuity loss, and boundary decoder await synchronization with collaborator code. No eligible MultiRep artifact has been received, so the manuscript makes no performance claim. Finally, a modular pose/track pipeline inherits missed detections and identity switches and may scale with the number of active people. The final study must quantify each limita-

tion through oracle/predicted tracks, corruption analysis, and population-scaling measurements. Until those gates close, the paper remains provisional and data-pending. MultiRep is synthetic, and single-person transfer cannot establish multi-person identity preservation; absent a real multi-person evaluation, any eventual conclusion must remain restricted to MultiRep and the declared controlled tracker corruptions.

Table 1. Planned MultiRep main results. Reproducible rows will show the mean and sample standard deviation over pre-registered seeds; official single-checkpoint rows will be marked. Protocol blocks answer different questions and will not be pooled into one ranking.

Method	Track protocol	Count supervision	Period-mAP↑	AP50↑	AP75↑	AvgMAE↓	AvgOBO↑
<i>Controlled oracle tracks</i>							
RepNet	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
TransRAC	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
<i>Controlled common predicted tracks</i>							
RepNet	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
TransRAC	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
<i>Native full-pipeline context; descriptive only</i>							
MultiCounter	Native	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	Native	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Table 2. Required protocol disclosure accompanying the main table. Native rows retain their official pipelines and are not pooled with common-input rows. Every field is manifest-bound.

Method	Result block	Input	Count labels	Period labels	Track source	Runs
RepNet	Common	PENDING	PENDING	PENDING	PENDING	PENDING
TransRAC	Common	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Common	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Common	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Common	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter	Native context	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	Native context	PENDING	PENDING	PENDING	PENDING	PENDING

Table 3. Planned construction-path diagnostic under oracle tracks. LT: local tempo; SR: soft routing; BS: boundary stitching; TC: track-continuity objective. Causal conclusions use removals from the full model and the factorial controls described below, not this order-dependent path alone.

Variant	LT	SR	BS	TC	Period-mAP↑	AvgMAE↓
Track-PAMS					PENDING	PENDING
+ LT	✓				PENDING	PENDING
+ SR	✓	✓			PENDING	PENDING
+ BS	✓	✓	✓		PENDING	PENDING
+ TC	✓	✓	✓	✓	PENDING	PENDING

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A. Extended Experimental Contract

This appendix records analyses that are required for the final evidence bundle but may move between main paper and supplementary material after results and the official CVPR 2027 format are frozen.

A.1. Artifact Manifest

Every table-eligible run must provide: model commit; evaluator commit; configuration and preprocessing fingerprints; seed; hardware and software environment; MultiRep release, source, download time, split, and checksums; per-video and per-identity ground truth and predictions; oracle/predicted-track status; logs; and generated-table provenance. An artifact lacking any required field is retained for diagnosis but cannot support a paper claim. UCFRep’s sealed test partition remains untouched until the method, checkpoints, and analysis code are frozen.

A.2. Mechanism Ablations

The mechanism study compares one global tempo per track against window-local tempo; uniform, hard, and soft routing; one, two, three, and five experts; PAMS independently applied to each track; and sliding-window PAMS with deterministic tempo bins. Boundary alternatives include non-overlapping windows, center-crop decoding, uniform response averaging, phase-aligned fusion, normalized overlap-add, and integer-window summation as a diagnostic failure mode. Training is compared with and without the synchronized within-track continuity term. An oracle router based on ground-truth periods may serve as an upper-bound diagnostic only when the required annotation exists, and must never be grouped with deployable methods. Arbitrary person-indexed parameter sets are excluded because identities do not persist across videos; efficiency instead compares shared batched execution with independent replicas of the same frozen checkpoint.

Table 4. Required characterization of the frozen MultiRep split before hypothesis testing. Every tempo or corruption stratum must also report its exact number of videos, people, and events.

Split	Videos	People	Events	Active IDs	Tempo-change	Pauses	Gaps	Window-eligible
Train	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Validation	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Test	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Table 5. Scope matrix for the three closest method families and this pre-results specification. “Intended” and “provisional” entries are not implementation or performance claims.

Method	Counting input	Output scope	Counting supervision	Tempo unit	Identity mechanism
PAMS [2]	Pose stream	Single stream	Self-supervised representation	Period-adaptive scales	None
MultiCounter [10]	RGB video	Person/events	Supervised MRAC	Per-person intervals	Joint association
MultiCounter+ [11]	RGB video	Person/events	Supervised MRAC	Long/short-aware	Dynamic Siamese query
Ours (specification)	Supplied pose tracks	Oracle person / pre-dicted track	Intended no person count/period	Window-local routing	Supplied track index

A.3. Robustness and Tracking Diagnostics

Natural tempo variation will be stratified by within-track cycle-duration dispersion. Deterministic count-preserving piecewise time warps, contiguous masking, and injected identity switches will measure degradation from each clean input using identical corruptions across methods. Perturbations are separated by pipeline level. Pre-tracking video corruptions rerun the frozen frontend and may change HOTA, IDF1, and IDSW; post-track pose/mask corruptions hold tracks fixed, so tracking metrics must remain unchanged. The protocol must record whether an identity switch changes only the track identifier or also the pose content. Results will be stratified by active-person count, asynchronous starts and stops, pauses, occlusion, and track fragmentation.

one failure case is mandatory; curves may not be manually redrawn.

A.4. Generalization and Efficiency

If protocol-compatible, a model trained on MultiRep will be evaluated without target-domain adaptation on the official RepCount and UCFRep single-person protocols. Dataset-native normalized MAE and OBO will retain their original names rather than being relabeled as MultiRep Avg-MAE/AvgOBO. Such transfer can support representation generalization only, not MRAC identity robustness. Efficiency will separate detector/pose/tracker preprocessing from counting latency and report parameters, FLOPs, peak memory, latency, counting-only throughput, and full-pipeline throughput as active identities increase. Shared batched execution and independent same-checkpoint replicas are timed separately. If ByteTrack [14] is selected as the common tracker, its exact code and checkpoint revision will be frozen in the manifest; this mention does not pre-select the frontend.

Qualitative figures must be generated directly from frozen per-frame artifacts and include persistent identity colors, valid masks, local period estimates, routing weights, fused responses, and ground-truth/predicted boundaries. At least

Table 6. Planned mechanism analysis. Parameter counts expose the cost of expert count and weight sharing.

Factor	Setting	Period-mAP $\uparrow$	AvgMAE $\downarrow$	AvgOBO $\uparrow$	Params (M) $\downarrow$
Tempo scope	Track-global	PENDING	PENDING	PENDING	PENDING
	Window-local	PENDING	PENDING	PENDING	PENDING
Router	Uniform	PENDING	PENDING	PENDING	PENDING
	Hard	PENDING	PENDING	PENDING	PENDING
Boundary	Soft	PENDING	PENDING	PENDING	PENDING
	Center crop	PENDING	PENDING	PENDING	PENDING
	Response average	PENDING	PENDING	PENDING	PENDING
	Normalized overlap-add	PENDING	PENDING	PENDING	PENDING

Table 7. Planned robustness report. Parenthesized values will be paired changes from the corresponding clean input with video-cluster bootstrap intervals.

Condition	Level	Severity	Stratum n	Period-mAP $\uparrow$	AvgMAE $\downarrow$	AvgOBO $\uparrow$	IDF1 $\uparrow$
Clean	–	–	PENDING	PENDING	PENDING	PENDING	PENDING
Natural tempo change	Data stratum	Low/medium/high	PENDING	PENDING	PENDING	PENDING	PENDING
Piecewise time warp	Post-track	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Contiguous occlusion	Pre-track	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Injected ID switches	Post-track	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Table 8. Planned efficiency report under one frozen hardware and timing protocol. The final generator expands every method into one row for each integer active-ID level from one through the frozen dataset maximum; the current Active IDs cell marks that pending grid rather than one aggregate measurement.

Method	Active IDs	Params (M) $\downarrow$	GFLOPs $\downarrow$	Peak GB $\downarrow$	Preprocess ms/frame $\downarrow$	Count ms/frame $\downarrow$	Pipeline FPS $\uparrow$
MultiCounter	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING